

CertainRAG Overview

CertainRAG is a lightweight Python library for runtime uncertainty estimation in Retrieval-Augmented Generation (RAG) pipelines. The library is designed to help developers evaluate how reliable a generated response is before presenting it to users.

Modern RAG systems often retrieve relevant context successfully but may still generate hallucinated, weakly grounded, or inconsistent answers. CertainRAG addresses this problem by combining multiple uncertainty signals into a single interpretable reliability estimate.

The library currently uses three complementary signals:

1. **Retrieval Confidence**
Measures how relevant the retrieved chunks are to the original query using normalized similarity scores from the developer's retriever or vector database.
2. **Faithfulness**
Uses a Natural Language Inference (NLI) model to estimate whether the generated answer is actually supported by the retrieved context.
3. **Semantic Consistency**
Measures whether multiple sampled responses remain semantically consistent across generations. Large variation between responses may indicate higher model uncertainty.

These signals are combined into a composite uncertainty score that can be used by downstream systems to make decisions such as:

- abstaining from answering,
- retrying retrieval,
- regenerating a response,
- or triggering additional tools.

CertainRAG is intentionally framework-agnostic. It does not perform retrieval or generation internally. Developers can integrate it with their existing pipelines built using LangChain, LlamaIndex, OpenAI APIs, local models, FAISS, Pinecone, Chroma, or other vector stores.

Example workflow:

User Query

- Retrieve Documents
- Generate Answer
- Evaluate with CertainRAG
- Decide System Action Based on Uncertainty

The library is designed primarily as an evaluation layer for reliability-aware AI systems.